

Jacinto™ Automotive Processor

Jacinto 7 Ethernet Switch

TI Confidential – NDA Restrictions

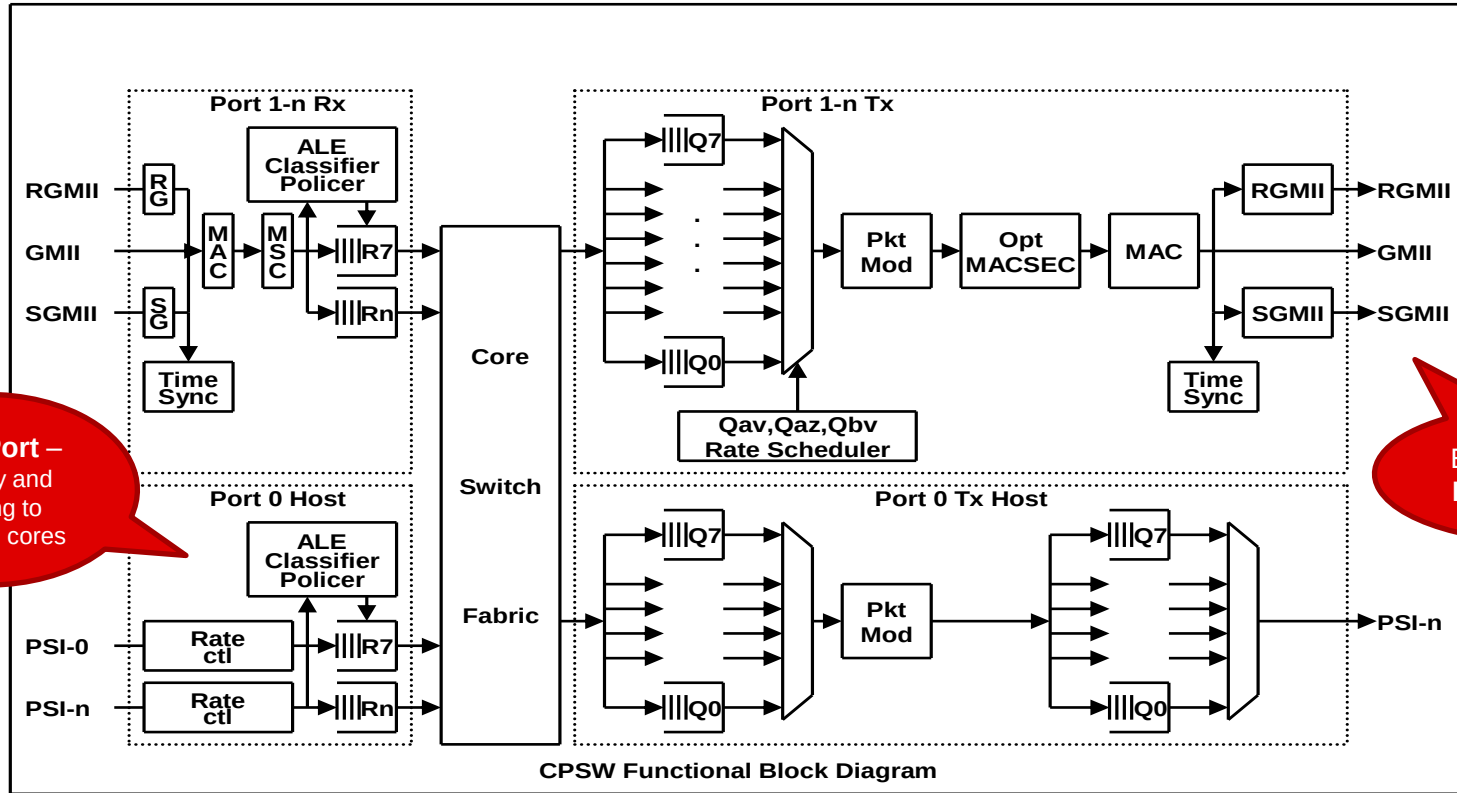
Agenda

- Ethernet Switch IP Features Overview
- Deep Dive – Software Architecture
 - CPSW LLD
 - Ethernet Firmware
- EthFw Demos
- Jacinto 7 EVM – CPSW support
- Deep Dive – EthFw
 - Folder Structure
 - Docs organization
 - Sample Examples
 - CPSW and PHY integration
 - Switch configuration & Debugging

IP Overview

ETHERNET SWITCH

Ethernet Switch – CPSW Functional Block Diagram



Host Port –
Classify and
Routing to
compute cores

**External MAC /
Ethernet Ports**

Ethernet Switch – Auto Use-cases entitlement

Security

- L2/L3 address whitelisting, L2 address blacklisting
- Advanced classification and Policing based on L2/L3 headers using Port number, Priority, ONU, DA, SA, VLAN inner/outer, EtherType, IPSA and IPDA
- Policing for rate limiting matched flows – three color marking on policed rates with drop controls
- Deep packet inspection via integrated compute cores

Safety Features

- SECDED ECC protection of table entries
- Ram error detection and correction (ECC) with full end to end packet protection (CRC)

AVB/TSN Features

- IEEE802.1Qav – Egress AVB rate shaping
- IEEE802.1Qbb and IEEE802.3x Flow control
- TSN Features
 - IEEE802.1Qbv/1Qaz time base scheduling
 - IEEE802.3br/IEEE802.1Qbu Frame Preemption

Ethernet Switch – Feature Summary

Key Building Blocks

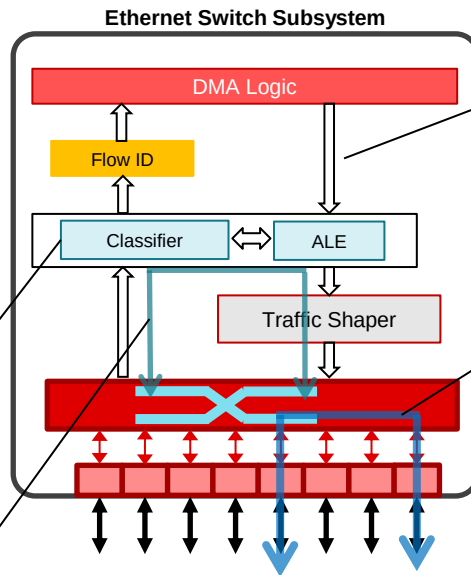
- ALE with 1k entries
- Classifier module for L2/L3 header inspection (96 HW classifier rules)
 - Ingress port: Rate-limit, filter
 - FlowID assignment for Host traffic
- AVB, TSN features support
 - Traffic shaper
- L2/L3 HW Checksum offload
- Inter-VLAN routing in HW

Traffic to Host DMA port

- **Classifier** for header inspection
- **FlowID** for packet steering and multiple queue support

Traffic to Host Inter-VLAN routing

- **ALE match**: traffic match to host port
- **Classifier**: Flow matching for HW offload
- **HW offload** for matched flow : packet header modification in HW



Transmit from Host

- **Classifier** support for rate-limit per flow
- **TSN support**: Time aware scheduling, pre-emption, Traffic Shaping with 8 HW Priority queue/port
- **Per-port** priority regeneration support

Forwarding Between Ports

- VLAN-aware line rate L2 switching
- **Per-flow** ingress port rate limiting, filtering
- **Per-port** broadcast, multicast rate limit

Firewall, Filter

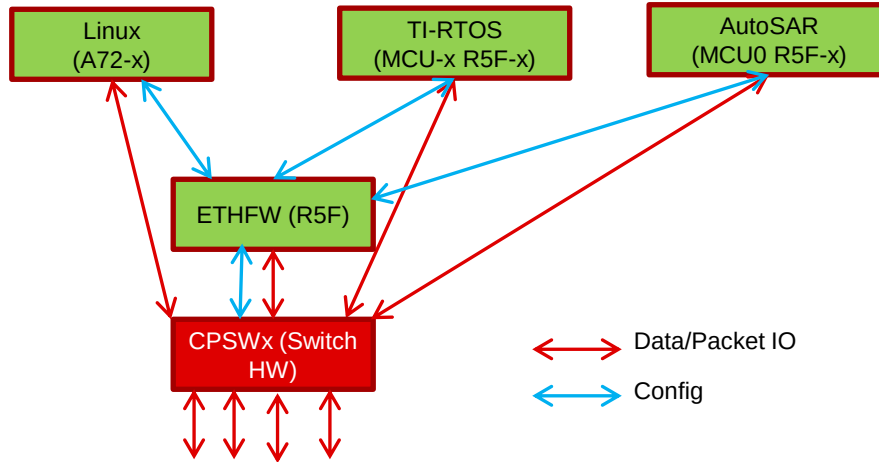
- L2-based filter (OUI deny, MAC authentication)
- Classifier based L2/L3 filtering
 - No match drop
 - MAC multicast range, IP CIDR mask filtering
- HW filter for untagged, IPV4, dual tag, IP fragment, L3 Next Header check

Deep Dive –Software Architecture

ETHERNET SWITCH

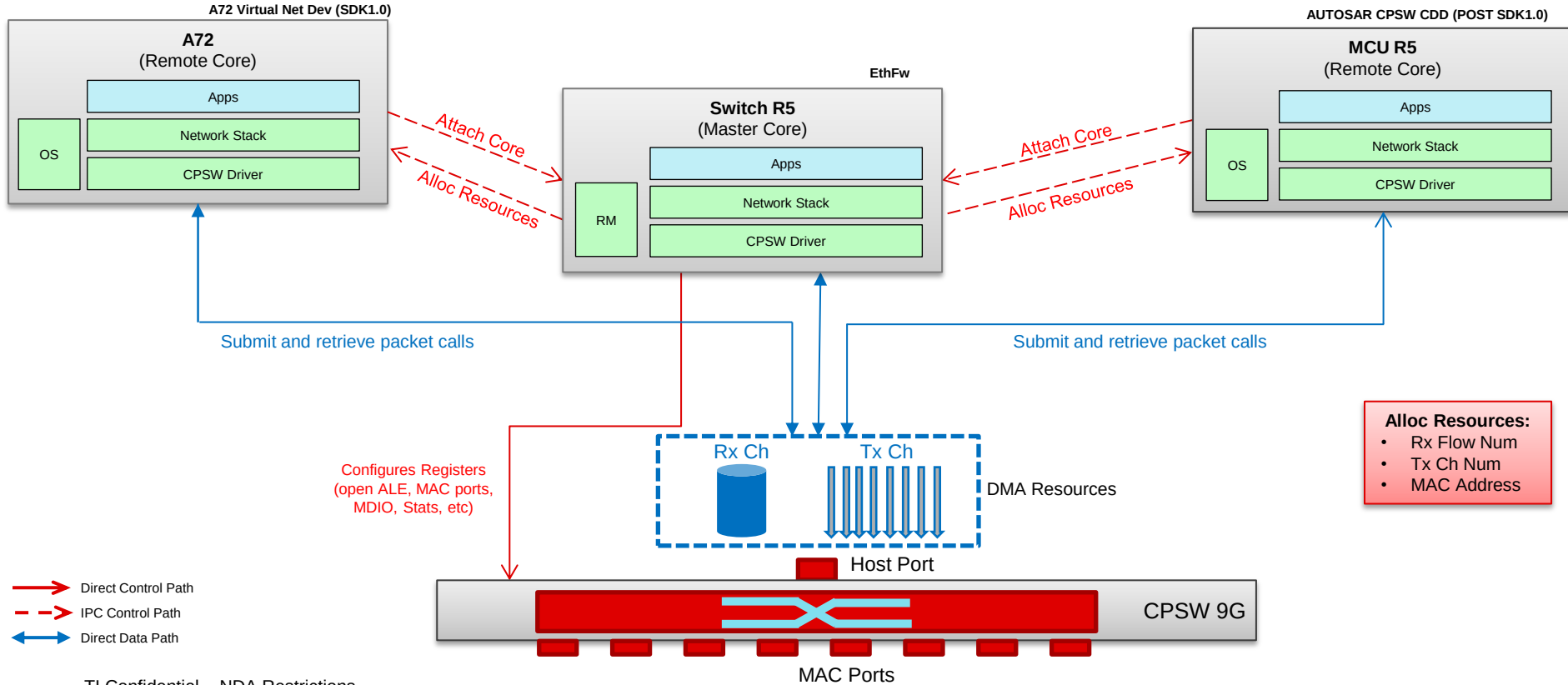
Ethernet Switch – Data Flow

Ethernet Firmware Supporting Multiple Clients



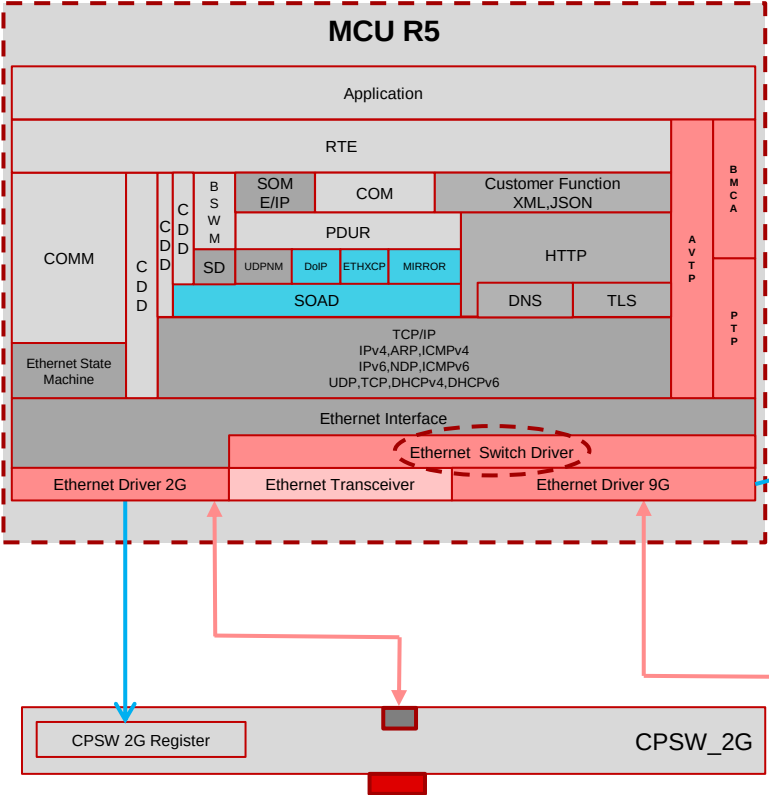
- Multiple CPUs on the SoC can perform network IO simultaneously using the switch once the switch is configured via the FW
- Subsequent packet IO can happen directly between the CPU specific network stack and switch HW via UDMA
- Switch configuration

Ethernet Switch – Data Flow Detailed

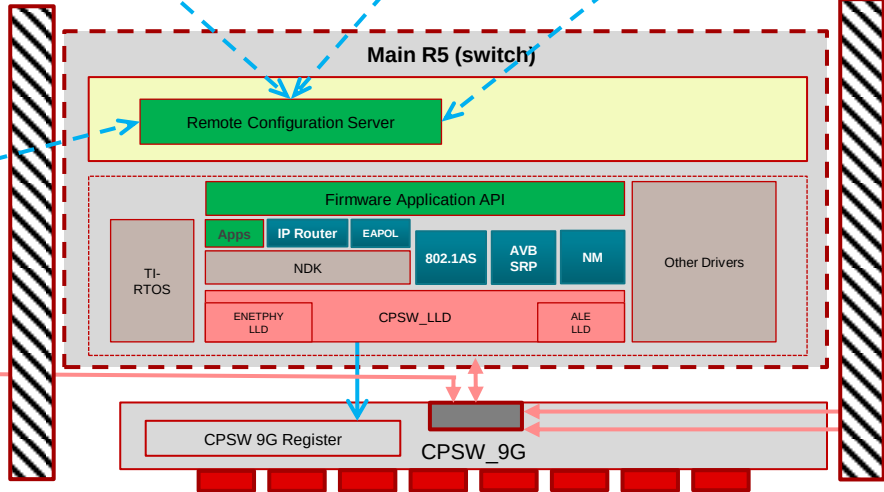
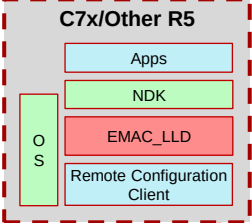
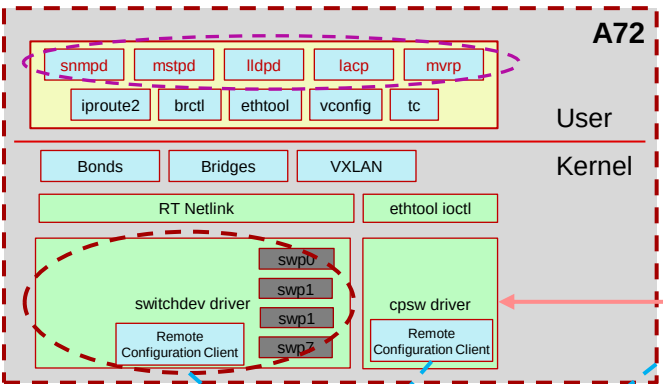


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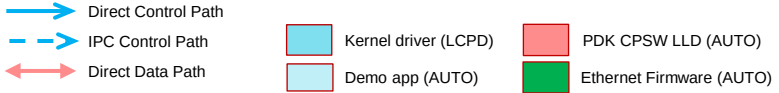
Ethernet Switch – Ref. AUTO Use-case



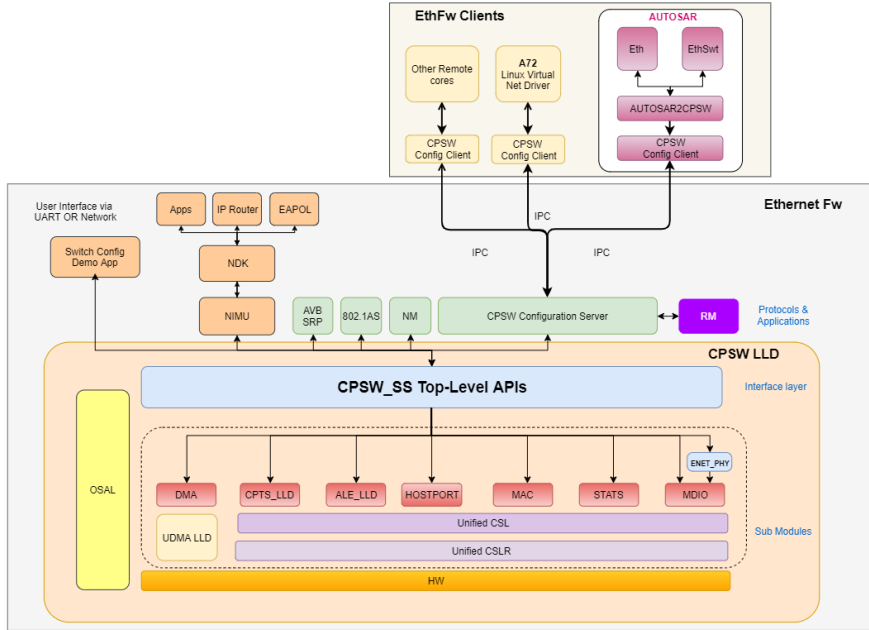
MCU Domain



Main Domain



Ethernet Switch – Software Stack



EthFw Software Stack

EthFw Submodules

- **CPSW LLD** – driver layer for CPSW9G IP
- **CPSW Remote Configuration Server**
 - CPSW Remote Configuration RTOS Client
- **Switch Resource Manager**
- **Switch Resident Protocols**
 - TCP/IP NDK
- **Switch Configuration App**

Software Releases

ETHERNET SWITCH

Ethernet Switch – EthFw 1.0: IS / IS NOT

SDK 1.0

- Basic Switching
- VLAN
- RGMII
- 1G/100M
- Host Port Packet Tx/Rx
- Switch R5 as packet source/sink
- IPC based switch configuration from other cores
- Linux virtual netdev driver integration
 - A72 data path
- TCP/IP stack (NDK)
- Multicast switching
- InterVLAN routing
- Network Security
- Classifier/Policer
- QoS/Packet Priority regeneration
- Traffic Shaping

Post SDK 1.0

- Timesync (802.1ASrev)
- Time aware scheduling (802.1Qbv) - EST
- SGMII
- RMII
- QSGMII and 2.5 Gbps support
- Port Mirroring
- Port Trunking
- IET (Interspersing express traffic)
- ENET PHY – power mgt
- Switch compliance testing
- Time sensitive networking

OS	CPSW 2G (Direct driver)	CPSW 9G (Virtual driver)
TI RTOS	Y	SDK 1.0
Linux	Y	SDK 1.0
AUTOSAR	Y	Post SDK 1.0
QNX	SDK 1.0	Post SDK 1.0

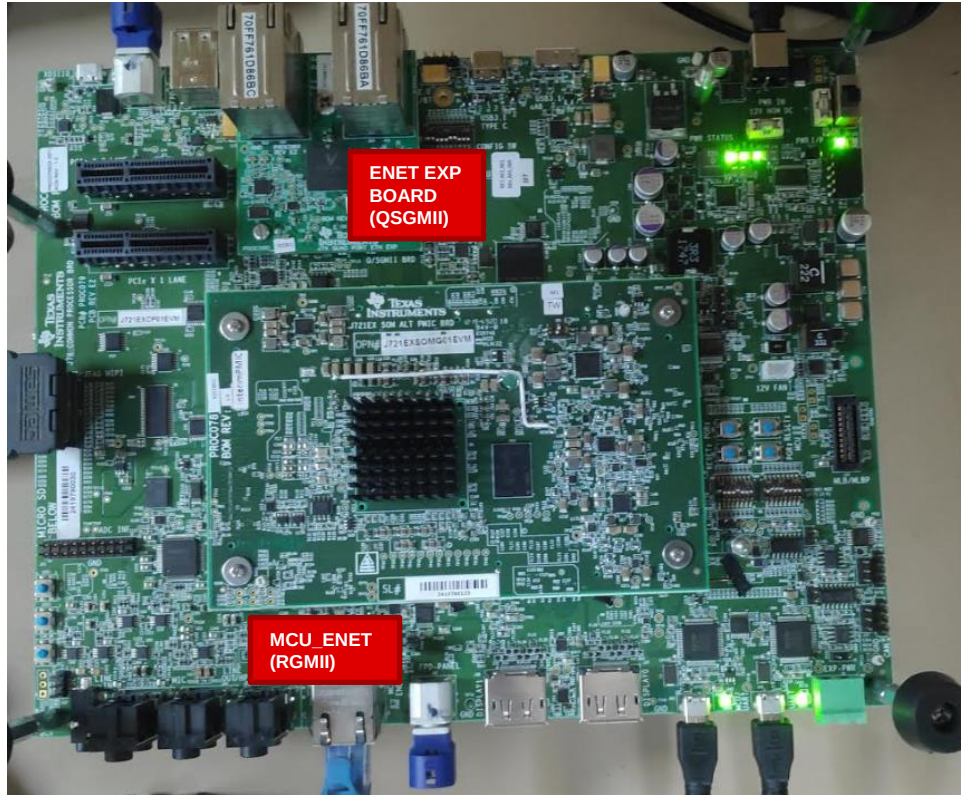
Jacinto 7 EVM: CPSW Support

ETHERNET SWITCH

Ethernet Switch – Jacinto 7 EVM Baseboard

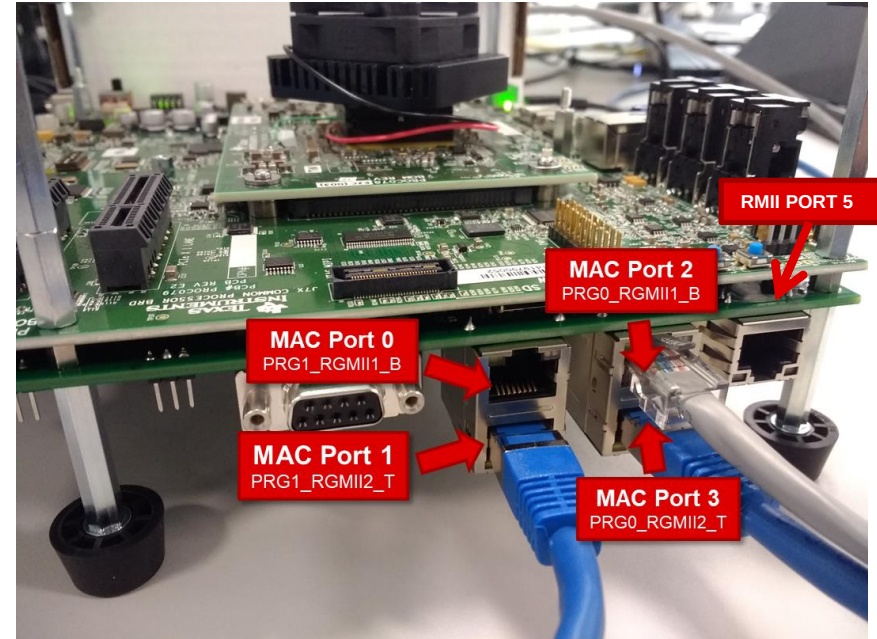
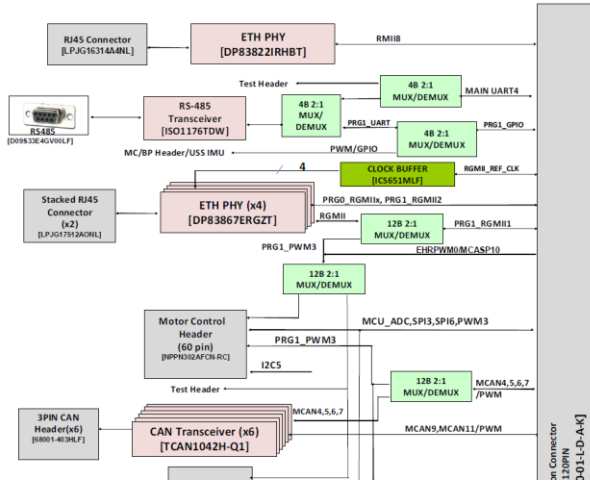
- **MCU_ENET**
 - DP83867 RGMII PHY
 - 1 Gbps
 - CPSW 2G MDIO bus
- **ENET_EXP DB**
 - 1 x QSGMII Expansion Daughter Board
 - Not supported as SDK1.0

With SDK1.0 only CPSW2G supported on Baseboard, GESI DB is needed for CPSW9G



Ethernet Switch – GESI Board

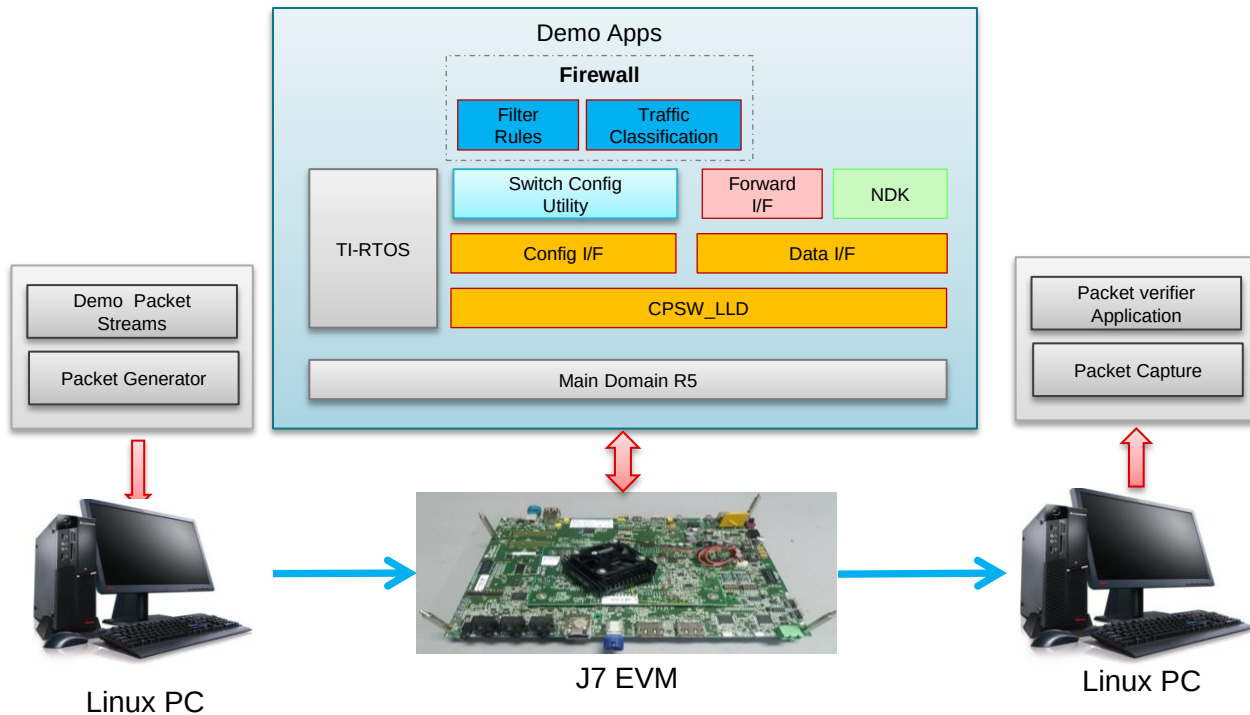
- 4 x RGMII PHYs
 - DP83867
 - 1 Gbps
 - CPSW 9G MDIO bus
- 1 x RMII PHY (Post SDK1.0)
 - DP83822
 - 100 Mbps
 - CPSW 9G MDIO bus



Demos

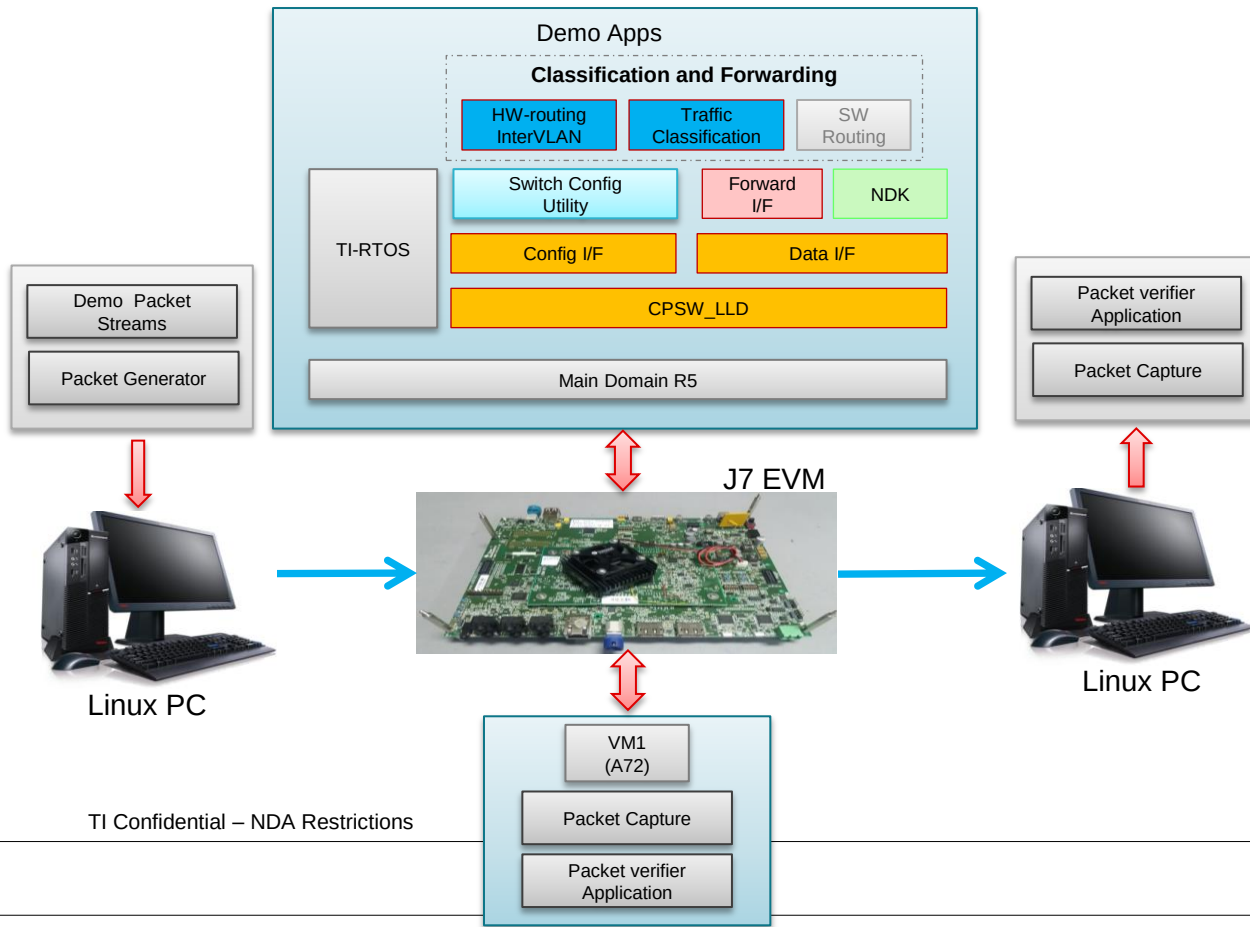
ETHERNET SWITCH

Ethernet Switch – Firewall



- **Demonstrate**
 - Packet classification Network security, firewall
 - Filter on L2 header: OUI, SA, VLAN
 - Filter on L3 header: frag, SA/DA
- **Differentiation**
 - Network security based on L2/L3 header fields (wire rate)
- **Timeline:** SDK 1.0

Ethernet Switch – Classification and InterVLAN Routing



- **Demonstrate**

- Packet classification
 - L2 header: addr, EtherType, VLAN, priority, port number
 - L3 header: SA/DA
- InterVLAN routing
 - Wire rate, up-to 4 routes

- **Differentiation**

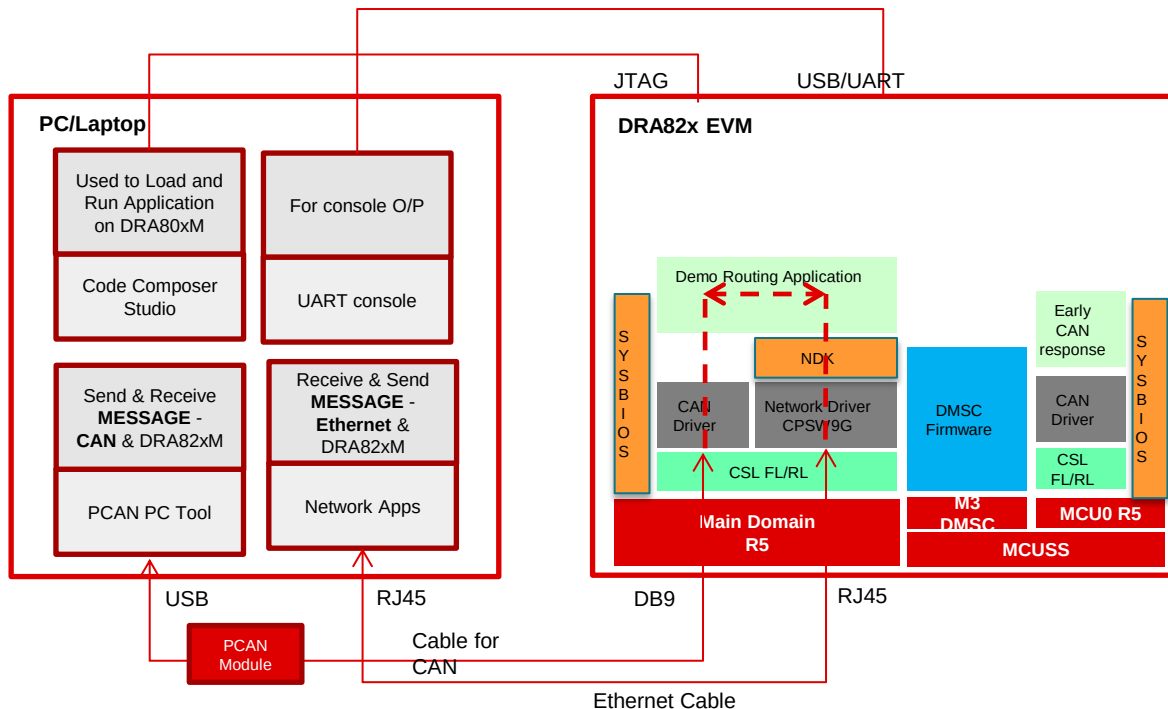
- Ethernet sharing across multiple cores / VMs
- HW packet header inspection/classification
- InterVLAN routing in HW at line rate

- **Timeline:** SDK 1.1

Gateway Demo: CAN ↔ Ethernet

- Demonstrates Latency and CPU Load for routing application
- Shows early CAN response
- AUTOSAR MCAL driver
- RTOS driver for CAN + Linux for ETH:

- Routing SW using just R5F – Cortex A free for other application usage
- <100us driver routing latency

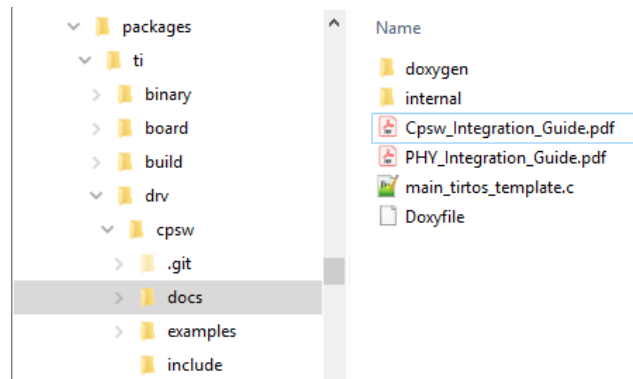


Deep Dive – Ethernet Firmware

ETHERNET SWITCH

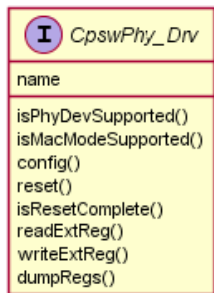
Ethernet Switch – CPSW Integration

- Integration into app:
 - CPSW configuration parameters: ALE, MAC, Host, MDIO
 - DMA configuration parameters
 - Print/tracing functions
 - CPSW init and open
 - Resource Manager attach
 - IOCTLs and helper functions
 - MAC/PHY configuration
 - Close sequence
- NDK integration C-file template
- CPSW LLD examples: loopback, sanity, NIMU



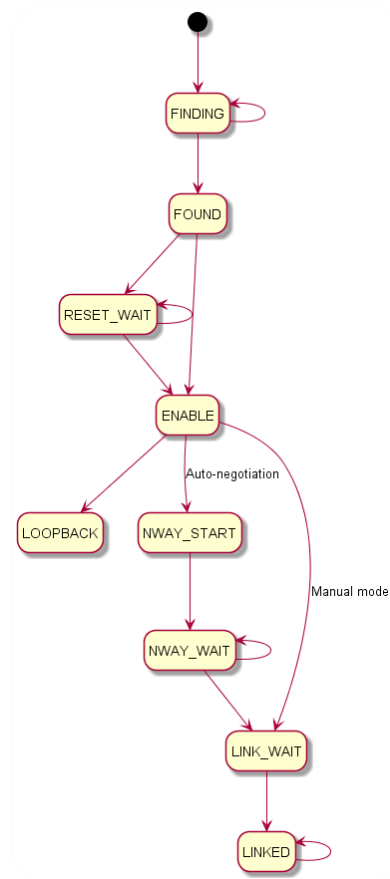
Ethernet Switch – PHY Integration

- PHY abstraction layer as part of CPSW LLD
 - State machine based
 - PHY-specific drivers can implement their own callbacks or reuse generic implementation



```
CpswPhy_Drv MyPhy_phyDrv =  
{  
    .name           = "My PHY driver",  
    .isPhyDevSupported = MyPhy_isPhyDevSupported,  
    .isMacModeSupported = MyPhy_isMacModeSupported,  
    .config          = MyPhy_config,  
    .reset            = MyPhy_reset,  
    .isResetComplete  = MyPhy_isResetComplete,  
    .runComplianceTest = NULL,  
    .readExtReg        = GenericPhy_readExtReg,  
    .writeExtReg        = GenericPhy_writeExtReg,  
    .dumpRegs          = MyPhy_dumpRegs,  
};
```

- Runtime device to driver mapping based on PHY ID
- Auto-negotiation and manual mode
- Clause-22 support
- “PHY Integration Guide” document provided as part of CPSW docs



Switch Configuration and Debugging

ETHERNET SWITCH

Ethernet Switch – Switch Configuration

- Default configuration loaded during EthFw init – part of
- Run-time switch configuration supported
Switch Config via Network (config GUI) & UART

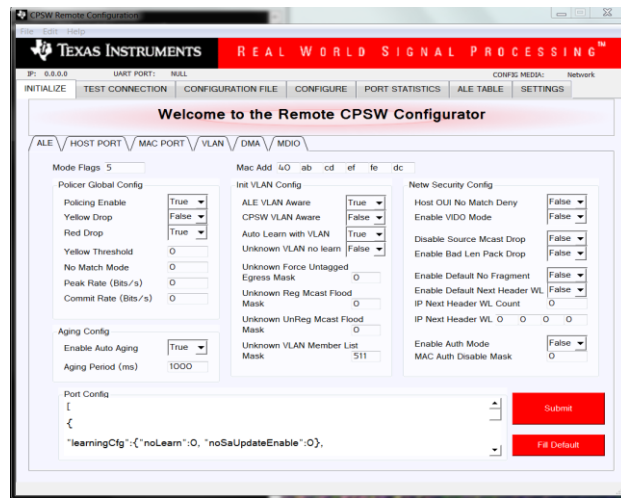


Fig – Switch Config GUI

```
=====
EthFw L2 Switching APP
=====
Host MAC address: 04:01:02:03:04:05
CPSW_9G Test on MAIN NAVSS
PHY 0 is alive
PHY 3 is alive
PHY 12 is alive
PHY 15 is alive
PHY 23 is alive
[NIMU_NDK] CPSW has been started successfully

CPSW NIMU application, IP address I/F 1: 192.168.1.108

=====
Switch Options
=====
1. Enable/Disable VLAN
2. Enable/Disable Multicast
3. Enable/Disable Rate Limiting
4. Enable/Disable InterVLAN
5. Print ALE & Policer Table
Enter your choice:
=====
```

Fig – UART Based Config

Ethernet Switch – Debugging

- Debugging Infrastructure
 - GEL Scripts
 - HW error stats report.
 - Dump Switch forwarding table
 - PHY reg read/write configuration
 - Debug and diagnostic APIs/IOCTLs
 - DMA statistics
 - Switch config dump IOCTLs
 - Runtime configurable trace support
 - GUI support to run stress tests, get switch statistics.

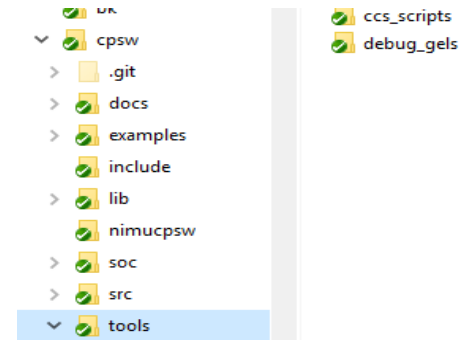
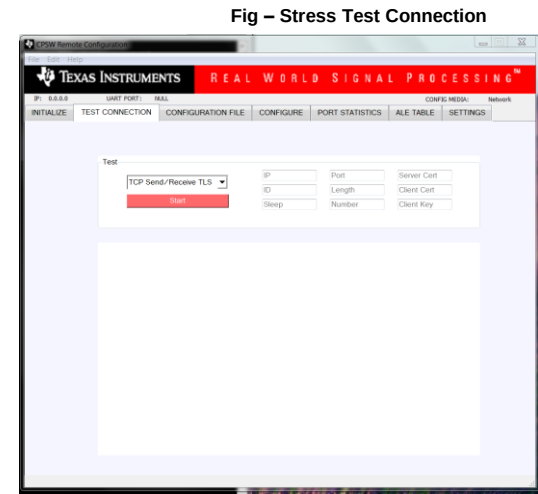


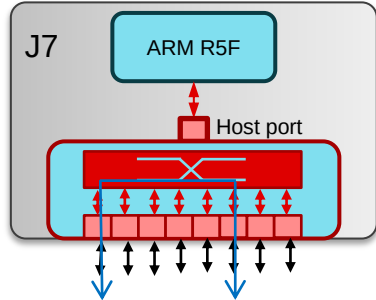
Fig – UART Based Config

Q & A

BACKUP

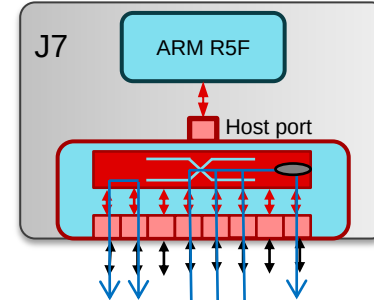
Ethernet Switch – Key Usecases

L2 Switching Usecase



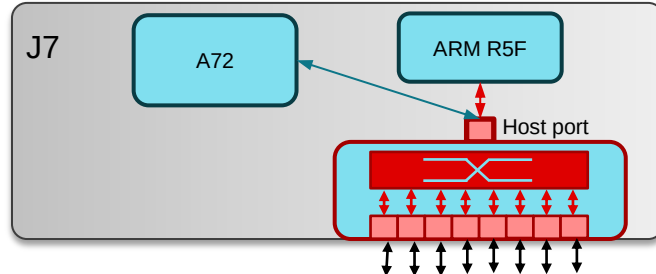
Traffic from any external port to any external port

L2 Aggregator Usecase



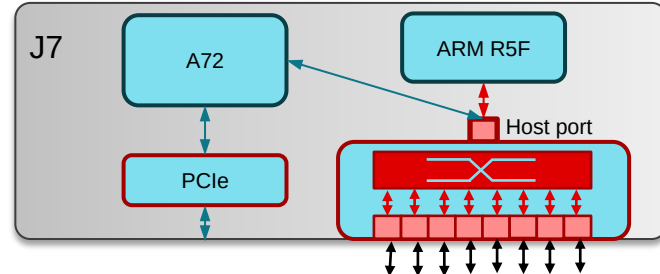
Traffic from multiple external ports aggregated and sent over external port

L2 Switching with Traffic to/from Compute Cores



Traffic from any external port to Jacinto 7 compute cores

PCIe / Ethernet Bridge



Ethernet Switch – Integrated Switch Key Value Propositions

Use-case	Key Features	Software Enablement
Gateway – Network Security	• IP address whitelisting • Protocol whitelisting • Rate limiting • Traffic shaping • MAC Authentication	• ALE driver with full IP entitlement • Support for switch configuration via AUTOSAR/Linux
IO Hub	• Support for 2-port driver & 9-port driver to connect vehicle network to switch	• Support for multicore data planes using packet based classifications (flows) • Single driver supporting all CPSW IP variants on Jacinto 7, Maxwell, Pascal. • Scatter-gather
Central Gateway	• Programmable L3 routing • InterVLAN routing (4 hardware routes) • 200ms Ethernet response	• AUTOSAR MCAL switch dev • LCPD Virtual netdev driver • TCP/IP stack (NDK integration) • Performance optimizations
Ethernet Surround View	• IET/EST • 802.1AS (PTP) • IEEE 802.1Qav (Forwarding)	• Time sync support • TSN support

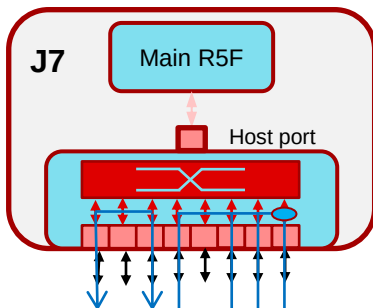
Ethernet Switch – Release Plan

Milestone	CPSW_LLD	EthFw	Demos	Date
SDK 0.8	• Basic Switching • VLAN • RGMII • 1G / 100M • Host TX / RX	• Switch config via UART		30 March 2019
SDK 0.9 Bring-up	• Multicast switching • InterVLAN routing • Ethernet PHY	• TCP/IP stack (NDK integration) • Switch config via telnet • A72 remote core support • Main R5_1 as packet source/sink • Bring-up complete on EVM	• CAN ↔ Ethernet gateway • L2 Switching – multicast, VLAN	30 June 2019
SDK 1.0 RTM	• Network Security • Classifier/Policer • QoS/Packet Priority regeneration • Traffic Shaping	• Linux A72 Virtual driver integration	• Firewall Demo • InterVLAN routing and Traffic Classification demo	30 Sep 2019

Ethernet Switch – Usecases Entitlement

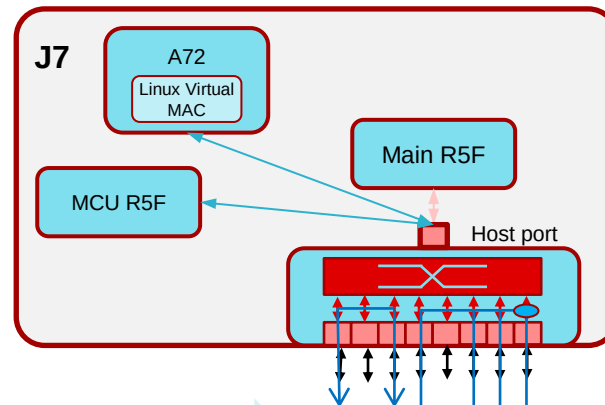
- Latency improvement to meet requirements
 - Direct data path from AUTOSAR R5 to Switch – eliminates need of IPC
 - Plan for TSN features to enable deterministic latency like EST, IET and switch FIFO tuning
- Optimized ETH driver for CPU load
 - Receive and transmit interrupt pacing
 - Support for scatter-gather to enable zero copy transmit
 - L3 checksum offload support
 - Support for enabling InterVLAN routing in hardware
- L3 routing
 - Dedicated switch R5 for Ethernet switch (CPSW9G) control – can be controlled via AUTOSAR or Linux
 - Multi-core support – support direct data path for each core (no IPC during data movement)
 - Support for enabling hardware features to classify and route traffic to intended core
 - Linux Virtual netdev driver & AUTOSAR Ethernet switch (CPSW9G) driver enablement

Ethernet Switch – Customer Enablement & Demos



L2 Switch Usecase

- Traffic from any external port to any external port
- Features:
 - Switch config via UART
 - VLAN
 - Multicast
 - RGMII, 100M / 1G



L2 Aggregator Usecase

- Traffic from multiple external ports aggregated and sent over external port
- Features:
 - Network security
 - Policer
 - QoS / Packet priority regeneration
 - InterVLAN routing

L2 Switching with internal cores

- Traffic from multiple external ports aggregated to J7 compute cores
- Features:
 - Traffic shaping
 - Traffic classification
 - TCP/IP (NDK) stack on Switch R5F
 - Packet TX/RX from MCU R5F

Performance

ETHERNET SWITCH

Ethernet Switch – Performance

Content to be added

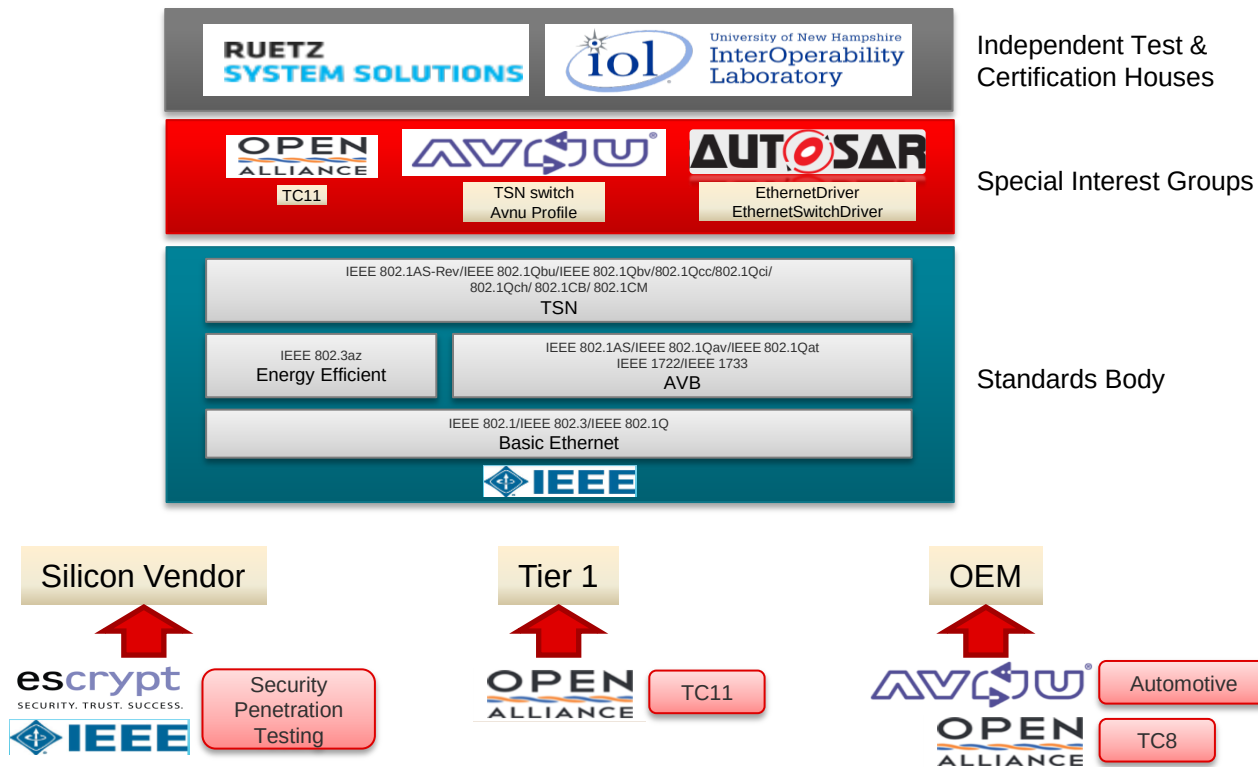
Ethernet Switch – Performance Measurement

Content to be added – CPU load, Network throughput

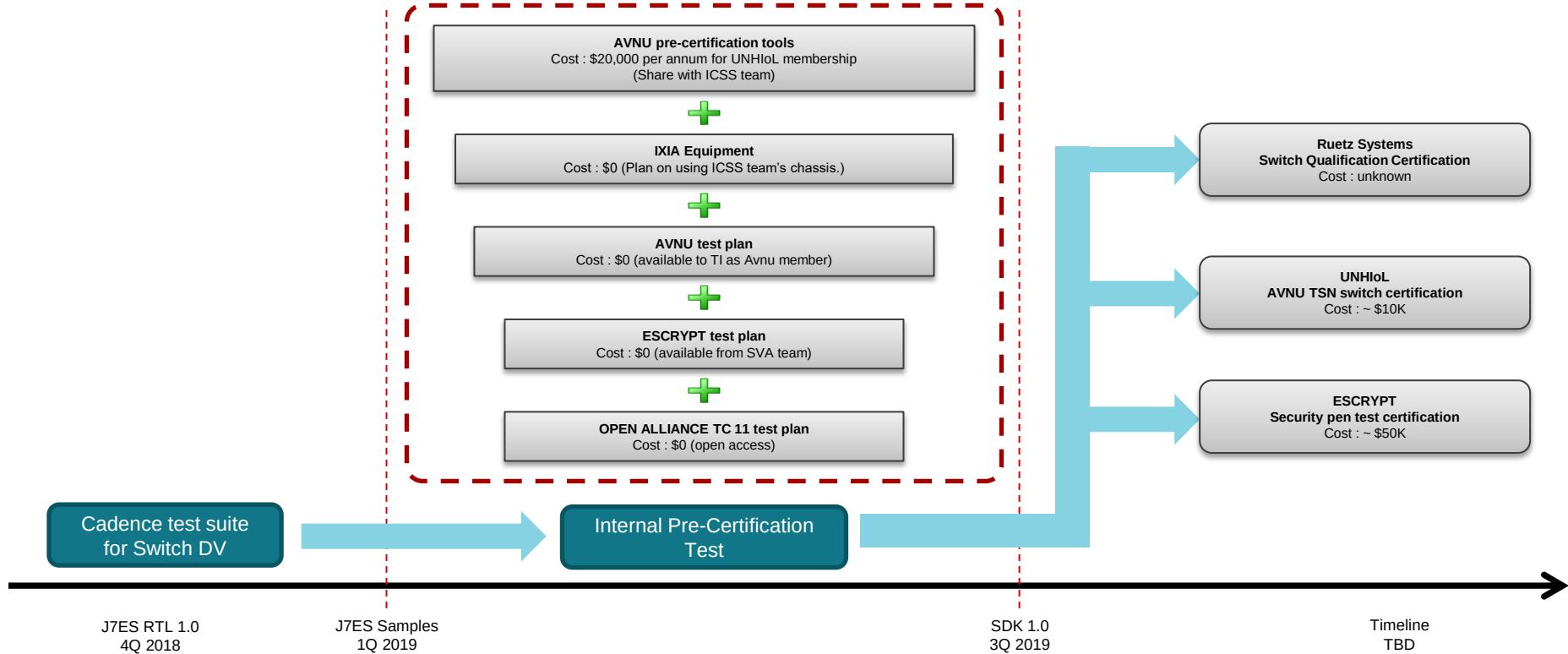
Switch Certification

ETHERNET SWITCH

Ethernet Switch – Standards Ecosystem



Ethernet Switch – Roadmap



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Ethernet Switch – Linux Virtual Net Driver